Mzuzu University

Department of Mathematics and Statistics

MATH 1201 Calculus

Test 1

Sunday, November 29, 2020 Time Allowed: 2 hours (02:00-04:00 PM)

INSTRUCTIONS

- *Write your name, registration number, programme and name of a lecturer in the spaces provided.*
- *Check that the paper has **9** questions.*
- *Attempt **all** questions.*
- *Write your answer in the space provided under each question.*
- *Show **all** your working in logical flow and with appropriate notation.*
- *Calculators are not allowed.*

Name :

Registration Number:

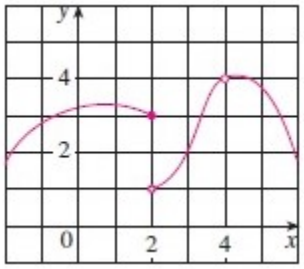
Program: :

Lecturer's Name: :

Question:	1	2	3	4	5	6	7	8	9	Total
Points:	8	9	3	5	6	5	4	4	6	50
Score:										

QUESTIONS

1. Use the graph of f to state the value of each quantity, if it exists. If it does not exist, explain why.



(a) Find $\lim_{x\to 2^-} f(x)$, $\lim_{x\to 2^+} f(x)$, and $\lim_{x\to 2} f(x)$. [3]

Solution: $\lim_{x\to 2^-} f(x) = 3$, $\lim_{x\to 2^+} f(x) = 1$, and $\lim_{x\to 2} f(x)$ does not exist since $\lim_{x\to 2^-} f(x) \neq \lim_{x\to 2^+} f(x)$

(b) Find $\lim_{x\to 4^-} f(x)$, $\lim_{x\to 4^+} f(x)$, and $\lim_{x\to 4} f(x)$. [3]

Solution: $\lim_{x\to 4^-} f(x) = 4$, $\lim_{x\to 4^+} f(x) = 4$, and $\lim_{x\to 4} f(x) = 4$

(c) Is f continuous at $x = 4$? Give an explain why it is (or not) continuous. [2]

Solution: f is continuous at $x = 4$ if $\lim_{x\to 4} f(x) = f(4)$. But $f(4)$ is undefined, so f is not continuous $x = 4$.

2. Evaluate the limit, if it exists;

(a) $\lim_{x\to 0} \sqrt{x^3 + x^2}\ \sin\frac{\pi}{x}$, [4]

Solution: Note that $-1 \leq \sin\frac{\pi}{x} \leq 1$ and $\sqrt{x^3 + x^2}$ is always positive when $-1 \leq x \leq 1$.
So $-\sqrt{x^3 + x^2} \leq \sqrt{x^3 + x^2}\sin\frac{\pi}{x} \leq \sqrt{x^3 + x^2}$.
Since $\lim_{x\to 0} \sqrt{x^3 + x^2} = \lim_{x\to 0}(-\sqrt{x^3 + x^2}) = 0$, then by the Squeeze Theorem $\lim_{x\to 0} \sqrt{x^3 + x^2}\sin\frac{\pi}{x} = 0$

(b) $\lim_{x\to\infty}(\sqrt{9x^2+x}-3x)$. [5]

Solution:

$$\begin{aligned}\lim_{x\to\infty}(\sqrt{9x^2+x}-3x) &= \lim_{x\to\infty}\sqrt{9x^2+x}-3x.\frac{\sqrt{9x^2+x}+3x}{\sqrt{9x^2+x}+3x}\\ &= \lim_{x\to\infty}\frac{(9x^2+x)-9x^2}{\sqrt{9x^2+x}+3x}\\ &= \lim_{x\to\infty}\frac{x}{\sqrt{9x^2+x}+3x}\\ &= \lim_{x\to\infty}\frac{\frac{x}{x}}{\sqrt{\frac{9x^2}{x^2}+\frac{x}{x}}+\frac{3x}{x}}\\ &= \frac{1}{\sqrt{9}+3}=\frac{1}{6}\end{aligned}$$

3. Find the value of the constant a that makes the function $f(x)$ continuous everywhere. [3]

$$f(x)=\begin{cases}1-3x, & \text{if } x<4\\ ax^2+2x-3, & \text{if } x\geq 4.\end{cases}$$

Solution: If a function is continuous it means the limit exist which implies that left hand sided limit is the same as right hand sided limit.

$$\begin{aligned}\lim_{x\to a^-}f(x) &= \lim_{x\to a^+}f(x),\\ \lim_{x\to 4^-}f(x) &= \lim_{x\to 4^+}f(x),\\ 1-12 &= 16a+8-3,\\ a &= -1.\end{aligned}$$

4. Find the derivative of $f(x)=x+\sqrt{x}$ using First Principles. [5]

Solution:

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{(x+h)+\sqrt{x+h}-x+\sqrt{x}}{h} \\
&= \lim_{h\to 0} \frac{(\sqrt{x+h}+h)-\sqrt{x}}{h} \cdot \frac{(\sqrt{x+h}+h)+\sqrt{x}}{(\sqrt{x+h}+h)+\sqrt{x}} \\
&= \lim_{h\to 0} \frac{x+h+2h\sqrt{x+h}+h^2-x}{h[(\sqrt{x+h}+h)+\sqrt{x}]} \\
&= \lim_{h\to 0} \frac{h+2h\sqrt{x+h}+h^2}{h[(\sqrt{x+h}+h)+\sqrt{x}]} \\
&= \lim_{h\to 0} \frac{1+2\sqrt{x+h}+h}{(\sqrt{x+h}+h)+\sqrt{x}} \\
&= \lim_{h\to 0} \frac{1+2\sqrt{x}}{\sqrt{x}+\sqrt{x}} \\
&= \frac{1+2\sqrt{x}}{2\sqrt{x}} \\
&= 1+\frac{1}{2\sqrt{x}}
\end{aligned}$$

5. Find the derivative of $y = \dfrac{x^2 \cos 3x}{\sqrt{x}}$. [6]

Solution:

$$\begin{aligned}
y' &= \frac{\sqrt{x}(x^2\cos 3x)' - x^2\cos 3x(\sqrt{x})'}{x} \\
&= \frac{\sqrt{x}[x^2(-3\sin 3x) + 2x\cos 3x] - x^2\cos 3x(\frac{1}{2\sqrt{x}})}{x} \\
&= \sqrt{x}[x(-3\sin 3x) + 2\cos 3x] - \frac{x\cos 3x}{2\sqrt{x}} \\
&= -3x^{\frac{3}{2}}\sin 3x + 2\sqrt{x}\cos 3x - \frac{x\cos 3x}{2\sqrt{x}} \\
&= -3x^{\frac{3}{2}}\sin 3x + \left(2\sqrt{x} - \frac{x}{2\sqrt{x}}\right)\cos 3x \\
&= -3x^{\frac{3}{2}}\sin 3x + \frac{3x}{2\sqrt{x}}\cos 3x \\
&= \frac{3x\cos 3x - 6x^2\sin 3x}{2\sqrt{x}} \\
&= \frac{3x(\cos 3x - 2x\sin 3x)}{2\sqrt{x}} \\
&= \frac{3}{2}\sqrt{x}(\cos 3x - 2x\sin 3x)
\end{aligned}$$

6. Find $\frac{dy}{dx}$ if $\sqrt{x+y} = 1 + x^2y^2$. [5]

Solution: $\frac{1}{2}(x+y)^{-\frac{1}{2}}(1+\frac{dy}{dx}) = 2xy^2 + 2x^2y\frac{dy}{dx}$ then $\frac{dy}{dx} = \frac{\frac{1}{2}(x+y)^{-\frac{1}{2}} - 2xy^2}{\frac{1}{2}(x+y)^{-\frac{1}{2}} - 2x^2y}$

7. If $h(2) = 2$ and $h'(2) = -3$, find [4]

$$\left.\frac{d}{dx}\left(\frac{h(x)}{x}\right)\right|_{x=2}.$$

Solution: We have $\frac{d}{dx}\left(\frac{h(x)}{x}\right) = \frac{xh'(x) - h(x)}{x^2}$. So

$$\left.\frac{d}{dx}\left(\frac{h(x)}{x}\right)\right|_{x=2} = \frac{2h'(2) - h(2)}{2^2} = \frac{2(-3) - 2}{4} = -\frac{8}{4} = -2$$

8. Find the equation of the tangent line to the curve $y = \sin(\sin x)$ at the point $(\pi, 0)$. [4]

Solution: $\frac{dy}{dx} = \cos x \cos(\sin x)$.
Slope of the tangent line is $m = \frac{dy}{dx}|_{x=\pi} = \cos\pi\cos(\sin\pi) = -1$. The equation of the tangent line is

$$\begin{aligned} y - y_1 &= m(x - x_1), \\ y - 0 &= -1(x - \pi), \\ y &= -x + \pi. \end{aligned}$$

9. For what values of a and b is the line $2x + y = b$ tangent to the parabola $y = ax^2$ when $x = 2$? [6]

Solution: We have $y' = 2ax$. So $y'|_{x=2} = 4a$. Note, from an equation of tangent line $2x + y = b$, that the slope is -2. That is, $y'|_{x=2} = 4a = -2 \Leftrightarrow a = -\frac{1}{2}$. It implies that $y = -\frac{1}{2}x^2$

So when $x = 2$ we have $y = -2$. Thus, an equation of tangent line is $y - (-2) = -2(x - 2) \Leftrightarrow y = -2x$. We conclude from this that $b = 0$.

END OF QUESTIONS